THE REMIX

A DiD Checklist

Walking through the Brazil mental health reform, step by step

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Glasgow · May 2026

Causal Inference: The Remix · Yale University Press · forthcoming 2026

Today’s example: the Brazil mental health reform

Dias & Fontes (2024)
“The Effects of a Large-Scale Mental Health Reform: Evidence from Brazil”
American Economic Journal: Economic Policy, 16(3): 257–289

What they study. Brazil’s 2002 psychiatric reform replaced large inpatient hospitals with *community mental health centers* (CAPS) — a major deinstitutionalization push.

Why we use it here. It is a clean staggered rollout, with a large panel of municipalities, and the right ingredients for a real DiD checklist.

The reform in one slide

What CAPS is. A *Centro de Atenção Psicossocial* — a community mental health center providing outpatient care, day programs, and psychosocial support. The reform created CAPS centers as a substitute for inpatient psychiatric hospitals.

Scale. Over 1,600 CAPS centers opened across 5,180 municipalities between 2002 and 2016. A municipality is either treated (gets a CAPS in some year) or never treated.

This is the canonical setup for staggered DiD: many units, different treatment dates, a long enough panel to see pre- and post-trends.

The headline finding

What worked.

- Increased **outpatient mental health care** use.
- **Reduced psychiatric hospitalizations**.

What didn't.

- ***Increased homicide rates*** — deinstitutionalization channel: vulnerable patients in the community without supervision.
- Hospital savings did not offset the social cost.

We'll focus on the homicide-rate outcome.

A checklist for staggered DiD

The Checklist Manifesto (Gawande, 2009) makes the case that even highly skilled experts skip steps under cognitive load. Pilots, surgeons, structural engineers — all use written checklists. Empiricists should too.

For staggered DiD, the steps are:

1. Define the target parameter.
2. Tabulate when units are treated.
3. Plot the rollout.
4. Plot the outcome by cohort.
5. Select covariates — with discipline.
6. Inspect the propensity score.
7. Estimate (CS, BJS, . . .).
8. Run an event study (universal baseline).
9. Sensitivity analysis (Rambachan–Roth).

Step 1: Define the target parameter

What is the question? Did opening a CAPS center, in the years after it opened, change the homicide rate in that municipality?

Three ingredients go into any target parameter:

1. A *difference in potential outcomes* $Y(1) - Y(0)$.
2. A *population* on which we average it.
3. A *weighting* rule $\mathbb{E}[\,\cdot\,]$.

Most disagreements about "the" DiD estimand are disagreements about which of these three you chose.

The target: ATT on the treated, in post-treatment periods

$$\text{ATT} \;=\; \mathbb{E}\Big[\, \underbrace{Y_{it}(1) - Y_{it}(0)}_{\text{(1) PO contrast}} \;\Big|\; \underbrace{D_i = 1,\ \text{Post}_t}_{\text{(2) population}} \,\Big]$$

In our case:

- **Contrast.** Y(1) = homicide rate with a CAPS; Y(0) = homicide rate without one.
- **Population.** Municipalities that *actually got* a CAPS, after they got it.
- **Weighting.** Group-time ATTs (g, t); we'll aggregate later.

Decide this *first*. Every downstream choice is meant to identify *this* parameter.

Step 2: Tabulate when units are treated

Before anything else, count the cohorts: how many municipalities got a CAPS in each year, and how many were never treated.

Why this table earns its place on the slide.

- It defines the **groups** (g) in the estimator.
- It tells you which cohort is the *smallest* — which sets a binding constraint on covariate count (see Step 5).
- It tells you which units are *always treated* (must be dropped) and *never treated* (the control group).

Next slide: the table itself.

CAPS introduction by cohort, 2003–2016

Cohort	Municipalities	Share of treated	Max cov. (EPV $\leq$ 7)
2003	47	3.5%	**6** (binds)
2004	70	5.2%	10
2005	95	7.1%	13
2006	**216**	**16.1%**	30 (largest)
2007	105	7.8%	15
2008	112	8.3%	16
2009	75	5.6%	10
2010	96	7.1%	13
2011	79	5.9%	11
2012	114	8.5%	16
2013	80	6.0%	11
2014	97	7.2%	13
2015	80	6.0%	11
2016	78	5.8%	11
Treated total	**1,344**	**100%**	—
Always-treated (2002, excluded)	296	—	—
Never-treated controls	3,836	—	—

2003 binds covariate count; 2006 dominates the simple-aggregation weight.

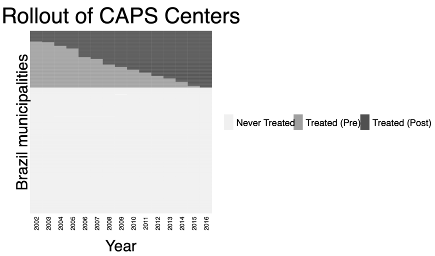
Step 3: Plot the rollout

Before any estimator, look at *when* treatment turns on, unit by unit.

Why look at it. The rollout picture tells you whether the staggering is rich enough to support staggered DiD (some units treated, some not, treatment dates spread across years), or whether you have a near-event-study problem (most units treated in one or two years).

Tool: `panelview` *(Yiqing Xu, Stanford). Rows are municipalities, columns are years, color encodes pre vs post treatment.*

Brazil CAPS rollout: 5,476 municipalities, 2002–2016



Light gray = never-treated. Medium gray = treated, pre-period. Dark gray = treated, post-period.

Step 4: Plot the outcome by cohort

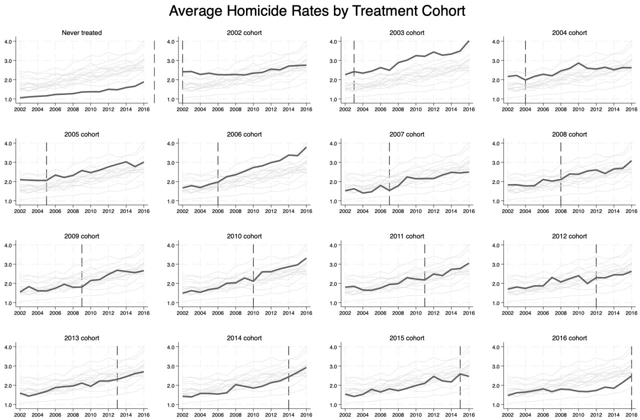
Now plot the *homicide rate* over time, separately for each cohort. Two purposes:

- See whether trends move in parallel across cohorts in the pre-period.
- Catch any anomaly — a missing year, a level break, an outlier cohort — before the estimator hides it from you.

One important reminder. DiD identifies $Y(0)$'s *trend*, not its *level*. Levels are differenced away. You are looking for parallel *movement*, not for cohorts that overlay each other.

This is the difference between DiD and synthetic control. Synth requires comparable levels and trends; DiD only needs comparable trends.

Homicide rate by treatment cohort



One small-multiples panel per cohort. Bold dark line = focal cohort. Dashed vertical = treatment year.

Step 5: Two ways to think about covariates

Path 1 — Outcome Regression (OR). Pick covariates that predict $Y(0)$'s *trend*. Ask: what are the ordinary drivers of homicide trends in Brazilian municipalities? Income, demographics, poverty, urbanization, federal transfers, education.

Path 2 — Propensity Score (PS). Pick covariates that predict *treatment timing*. Ask: who got a CAPS first? Larger cities? Poorer states? Earlier hospital capacity?

Doubly robust uses both. If you can correctly specify either the outcome trend or the propensity score, DR is consistent. You don't have to be right twice.

Use common sense first. Run LASSO or stepwise selection only after a domain-informed shortlist.

Step 5 (cont.): the 7-events-per-covariate rule

Peduzzi, Concato, Kemper, Holford & Feinstein (1996), *"A simulation study of the number of events per variable in logistic regression analysis,"* *Journal of Clinical Epidemiology*, 49(12): 1373–1379.

The rule. A logistic regression needs roughly 7–10 "events" (the smaller of the two outcome classes) per covariate to avoid overfit and unstable coefficients.

In a propensity score model for staggered DiD, the "event" is the number of treated units in a 2×2 comparison. If cohort g has 47 municipalities (our 2003 cohort), the EPV ceiling is roughly $\lfloor 47/7 \rfloor = 6$ covariates.

Different 2×2's can carry different covariate counts. The smallest cohort sets the binding limit.

Step 5 (cont.): standardize, for the computer's sake

The problem. Some covariates look like `poptotaltrend = population` $\times$ `year`. In Brazil that can be 150 million or larger. When the propensity-score logit tries to invert a matrix with entries that big, it doesn't fail like a textbook tells you — it fails like a computer.

The fix. Convert every continuous covariate to a *z-score*: subtract the mean, divide by the SD. Now everything is on the same scale, the matrix inverts cleanly, and your estimates don't shift just because someone changed units.

This is not econometrics. It is numerical hygiene. The model is the same; only the computer's life is easier.

Step 5 (cont.): the balance table the paper actually used

The published spec used **31 covariates** in the propensity score and outcome model. The balance table (baseline year 2002, excluding the always-treated 2002 cohort) reports a normalized difference for each one.

31 candidates included things like

- Log GDP per capita
- Federal cash-transfer receipts
- 14 age $\times$ sex population bins
- Rural population share
- Pre-period homicide trend
- Pre-period health-spending trend

Imbalance is real.

- 9 covariates with $|\text{norm. diff.}| > 0.25$
- Largest: `poptotaltrend` at $+0.69$
- Smallest cohort (2003): EPV cap of 6
- Largest cohort (2006): EPV cap of 30

31 candidates, 6 slots in the binding 2×2. Step 5 picks which 6.

Step 6: Inspect the propensity score

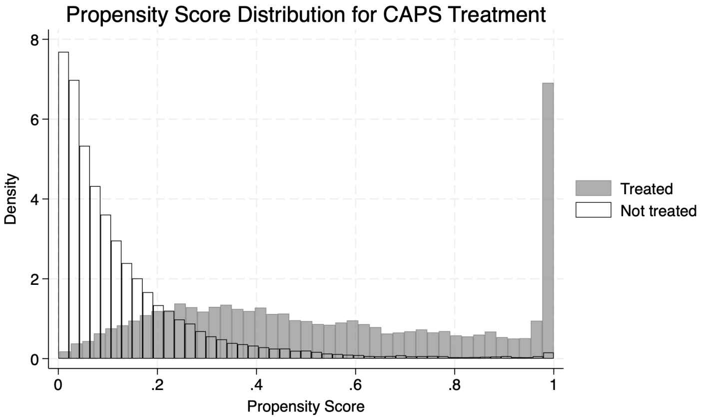
After picking covariates, fit `probit treat $controls if ano==2002` and plot $\widehat{p}(X)$ for treated and untreated municipalities.

What you want to see: substantial **overlap**. Treated and untreated propensity-score distributions should share most of their support. Mass at $\widehat{p} \approx 1$ for controls is a red flag — those are units the model thinks should have been treated but weren't.

Standard practice: trim controls with $\widehat{p} > 0.995$ before estimating. The `did` R package does this automatically.

Next slide: histogram from a fitted PS in our setup.

Propensity score distribution, by treatment status



Gray = treated. Black outline = never-treated. Mass in the right tail of controls is the trimming target.

Step 7: Estimate with the Callaway–Sant'Anna estimator

Three choices, made deliberately:

- **Baseline period.** Use `long2` in Stata / `base_period = "universal"` in R. This pins every event-time effect to $t = -1$, so pre-trend coefficients are real falsification tests.
- **Estimation method.** `dr` (doubly robust) or `ipw` or `reg`. DR if you trust neither model individually; OR if you don't want to extrapolate from a logistic regression with overlap problems.
- **Control group.** `notyet` — not-yet-treated comparison. Bigger sample than never-treated only; no pre-trend restriction.

Step 7 (cont.): the aggregation question

`csdid` returns group-time ATTs: $\widehat{\text{ATT}}(g,t)$ for every cohort g and post-period t. Aggregating these is a choice with consequences.

- **Simple ATT.** Average over all (g,t) post-treatment cells, weighted by cohort size $\times$ exposure length.
- **Group ATT.** First average within cohort to one number per g; then average across cohorts.
- **Event-study ATT.** Average across cohorts at each event time $e = t - g$.

Big cohorts have big weights. The 2006 cohort (216 units, 16% of treated) will dominate any size-weighted average.

Step 7 (cont.): a small artifact, to make weighting concrete

Two cohorts. **Group A**: 5 post-period cells, 500 units each. **Group B**: 2 post-period cells, 100 units each.

cell	Group A (n = 500)					Group B (n = 100)	
	A_1	A_2	A_3	A_4	A_5	B_1	B_2
$\widehat{ATT}$	0.10	0.12	0.14	0.11	0.13	0.50	0.60

Both aggregations weight by population — just at different units:

- **Simple** (each *cell* weighted by N_g): $\frac{5 \cdot 500 \cdot 0.12 + 2 \cdot 100 \cdot 0.55}{5(500)+2(100)} = \frac{300+110}{2700} =$ **0.152**
- **Group** (each *cohort* weighted by N_g): $\frac{500 \cdot 0.12 + 100 \cdot 0.55}{600} = \frac{60+55}{600} =$ **0.192**

Simple amplifies long-exposure cohorts; group neutralizes exposure length.

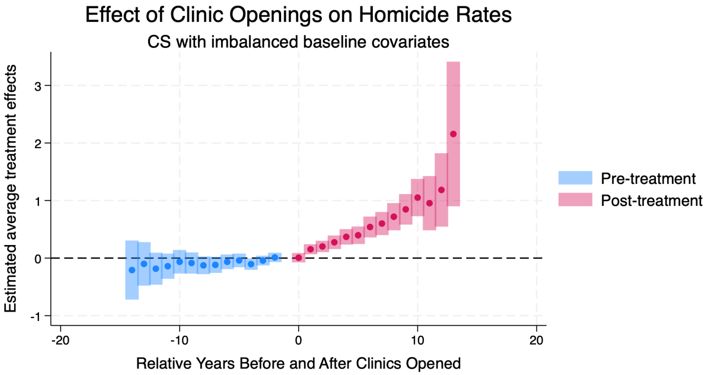
Step 8: The event study, with a universal baseline

The event study reports $\widehat{\text{ATT}}(e)$ for each event time $e = t - g$. Two reasons we use a *universal* baseline ($t = -1$ pinned to zero):

- **Pre-trend coefficients become falsification tests.** Each pre-period $e < 0$ is now a "long difference" relative to $t = -1$. The identifying assumption is that *post-period* long differences in $Y(0)$ are zero; pre-period ones should be too.
- **Post-period coefficients become interpretable on the same scale.** Every $\widehat{\text{ATT}}(e)$ for $e \geq 0$ is the effect of an additional $e + 1$ years of treatment exposure relative to the year before.

In Stata: `csdid ... long2`*. In R:* `base_period = "universal"` *inside* `att_gt()`*.*

The event study: CAPS effect on homicide rates



Note: Uses the 2008–2010 eventually treated and the never treated counties as controls. Circles are ATT(l) estimates by relative time, l. All groups and bands are 95% uniform confidence intervals. Mean homicide rate was approximately 1.23 in 2002.

Years relative to CAPS opening. Bars are 95% uniform confidence intervals. Pre-trend points near zero pass the falsification test.

Step 9: Sensitivity — when pre-trends are imperfect

Pre-trends rarely look *exactly* flat. The question is then: how large would the parallel-trends violation have to be for our conclusion to flip?

Rambachan & Roth (2023), *"A More Credible Approach to Parallel Trends," Review of Economic Studies*, 90(5): 2555–2591.

The idea. Treat the magnitude of post-treatment parallel-trends violation as bounded by some multiple M of the largest pre-period violation. For each M, compute the *robust* confidence interval for the post-treatment effect. Then plot the CI as M grows.

The estimator is no longer point-identified — it's a set. Honest about what the data can and cannot rule out.

Step 9 (cont.): the code

The Stata implementation is the `honestdid` package (Rambachan, Roth, Wing). Two steps:

```
* 1. Estimate the CS event study with long2 / universal baseline
csdid2 homicide_rate $controls, gvar(g) ivar(cod) ///
    time(ano) long2 method(drimp) notyet
estat event, window(-4 5)

* 2. Run the sensitivity analysis on the post-period average
matrix l_vec = J(1, 1, 1) // weight on Post_avg
honestdid, pre(3/5) post(2/2) ///
    mvec(0(0.25)2) delta(rm) ///
    l_vec(l_vec) b(e(b)) vcov(e(V))
```

- `mvec(0(0.25)2)`: sweep M from 0 to 2 in steps of 0.25.
- `delta(rm)`: relative-magnitudes restriction (the standard choice).

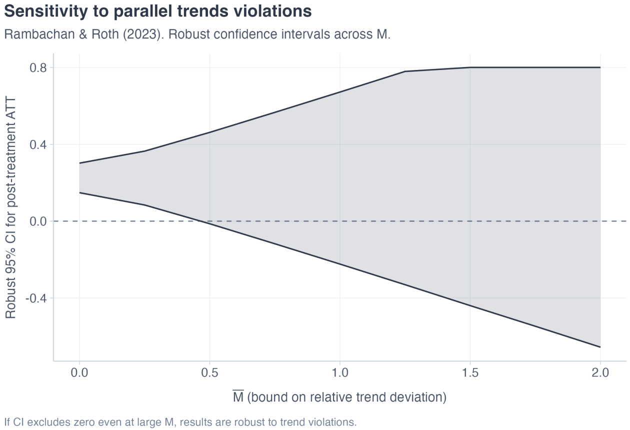
Reading the HonestDID plot

The output is a forest of confidence intervals, one for each M.

- $M = 0$: the original CS estimate.
- $M = 0.5$: PT violation half as large as worst pre-period.
- $M = 1$: as large as worst pre-period.
- $M = 2$: twice as large.

Breakdown point: the M at which the CI just touches zero. Surviving $M \geq 1$ means robust to anything seen in the pre-period.

The HonestDID plot for CAPS



Each bar is a 95% robust CI for the post-treatment ATT at a value of M. Black diamond at $M = 0$ is the original CS estimate.

A counterpoint: “Don’t Do DiD”

Guido Imbens (Stanford, Nobel 2021) recently gave a talk titled *“Don’t Do Difference-in-Differences”* — DDDiD.

His argument. Conditional parallel trends is a strong, untestable assumption. If you don’t believe it, an estimator that requires it is the wrong tool. He recommends other panel methods — chief among them **synthetic control** and **synthetic DiD**.

Imbens, `youtube.com/watch?v=GG627T4GbqU`

Why he might be right — and the 1970s caution

The empirical crisis of the 1970s was not a failure of econometrics. The math of 2SLS, regression, and instrumental variables is fine. Smart people could prove these results standing on their heads.

What failed was the marriage of method to data. Orley Ashenfelter and Janet Currie recall a time when researchers wouldn't even *disclose what their instrument was*. Identification was assumed, not defended.

The more complex the estimator, the higher the returns to knowing your data. And the higher the cost of being wrong about its assumptions.

But beware absolutes

Many people will tell you that **including covariates in a DiD is always suspect.** Or that **you should never use TWFE under staggered adoption.** Or that **conditional parallel trends is too strong to defend.**

Counter-example. In the LaLonde–Dehejia–Wahba data, we just saw that conditional-on-X DiD with the right covariates *does* recover an estimate close to the RCT benchmark. Proof by contradiction.

Don't throw rocks in glass houses. The smart move is to know your data well enough to know when conditional PT might hold, when it might not, and to argue the case in both directions.

Tomorrow: when you don't believe conditional PT

If you have reason to doubt conditional parallel trends — and you can't shore it up with covariates, sensitivity analysis, or design — then you should be looking at a *different* class of estimator.

Tomorrow we cover:

- **Synthetic control** (Abadie, Diamond & Hainmueller).
- **Synthetic DiD** (Arkhangelsky, Athey, Hirshberg, Imbens & Wager).

These trade one set of assumptions for another — and that trade is the conversation worth having.